Shape of cubics

1 Yes

2

a they become very large in the positive direction

b they become very large in the negative direction

3 Odd, because the ends of the function point in opposite directions.

4

a $x < 0$ **b** $x > 0$

c $(0, 0)$ **d** The point of inflection

5

a $x > -3$ **b** $x < -3$ **c** $(-3, 0)$

6 y increases more slowly on $y = \frac{1}{2}x^3$ than on $y = x^3$

7 $x = -1, \quad x \geq 3$

8

a $(0, 0)$ **b** $(0, 3)$ **c** $(0, 4)$

Finding intercepts and equations of cubics

9

a Negative **b** $(0, 3)$ **c** $x < 0$ **d** $x > 0$

e $(0, 3)$

10

a No **b** No **c** No **d** Yes

11

a $x = -2$ **b** $y = 4$

12 3

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

13

a negative **b** $(0, 3)$ **c** **A**

14 $y = (x - 10)^3$

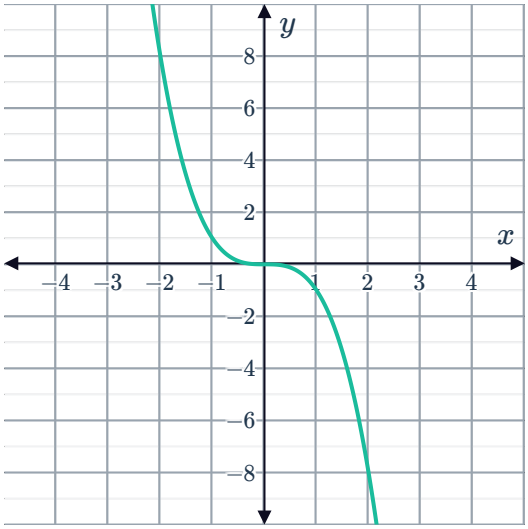
Graph of cubics

15

a

x	-2	-1	0	1	2
y	8	1	0	-1	-8

b

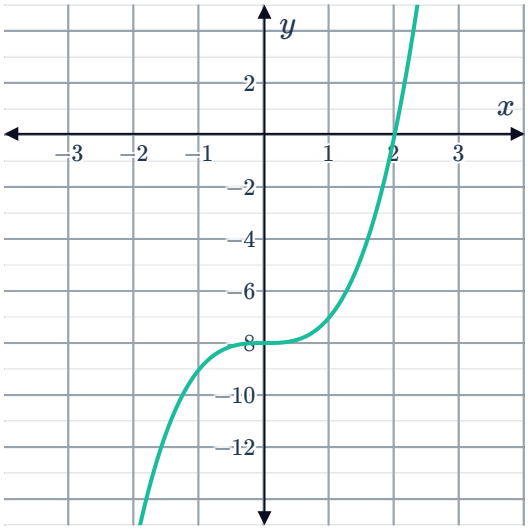


16

a $x = 2$ **b** $y = -8$

c $(0, -8)$

d



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

17

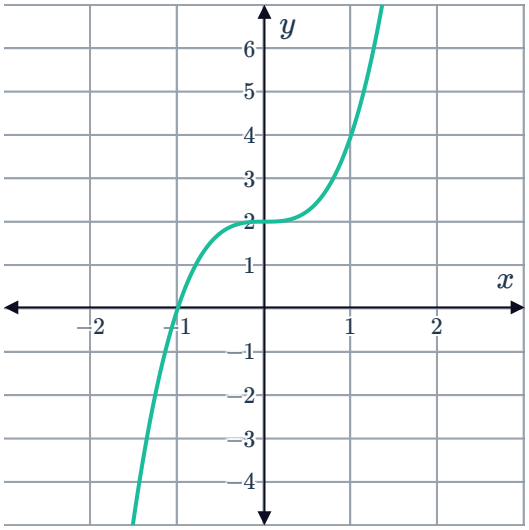
a

i Increasing

ii More steep

iii $(0, 2)$

iv



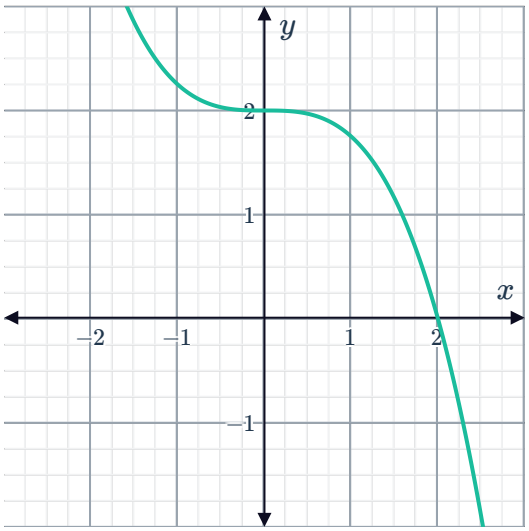
b

i Decreasing

ii Less steep

iii $(0, 2)$

iv



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

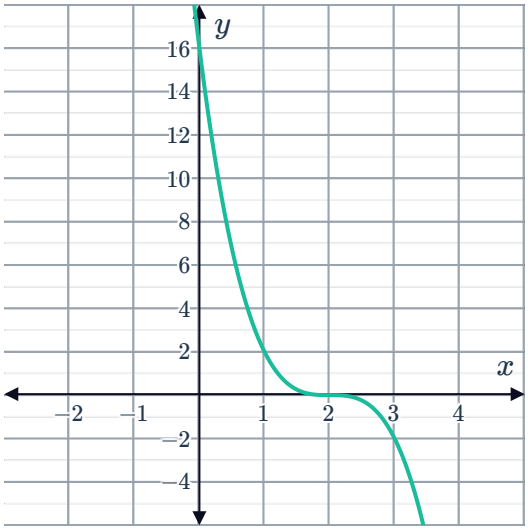
c

i Decreasing

ii More steep

iii $(2, 0)$

iv



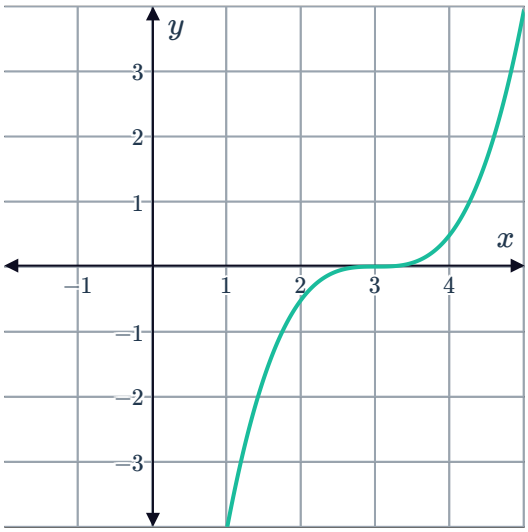
d

i Increasing

ii Less steep

iii $(3, 0)$

iv



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

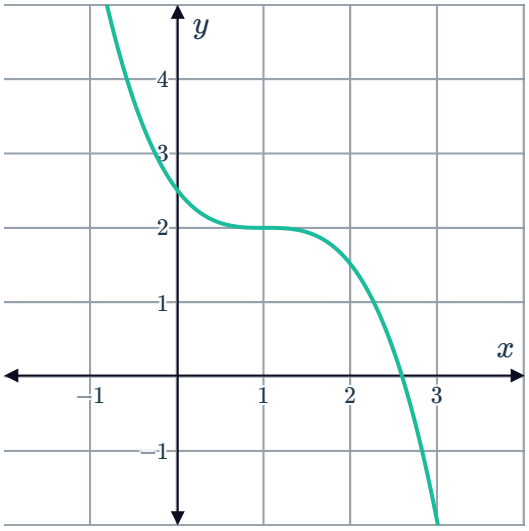
e

i Decreasing

ii Less steep

iii $(1, 2)$

iv



18

a $x = 2$

b $y = 6$

c $(1, 3)$

d

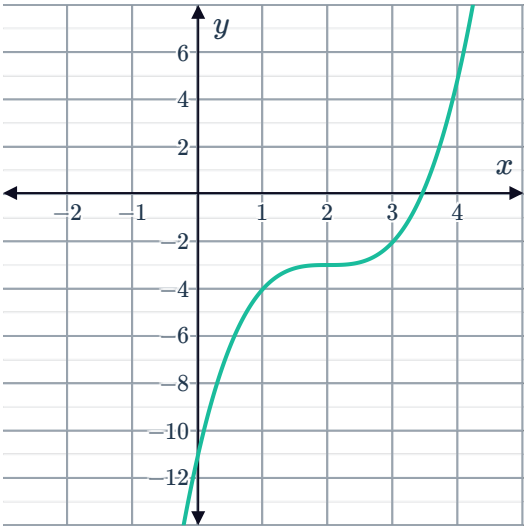


© ClipSAT: By Mr. Mohamed Abdallah +966597688647

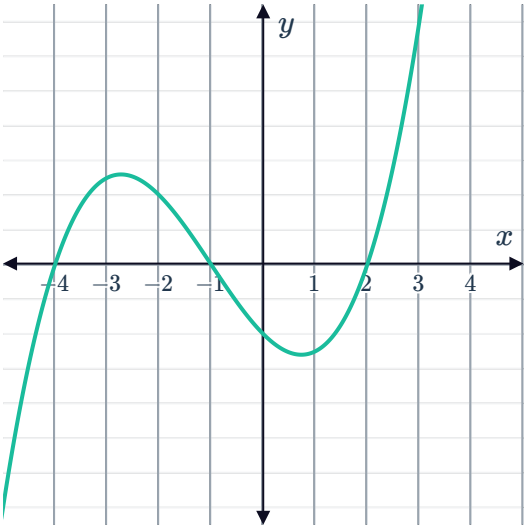
19

a Move the graph to the right by 2 units and down by 3 units.

b



20



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

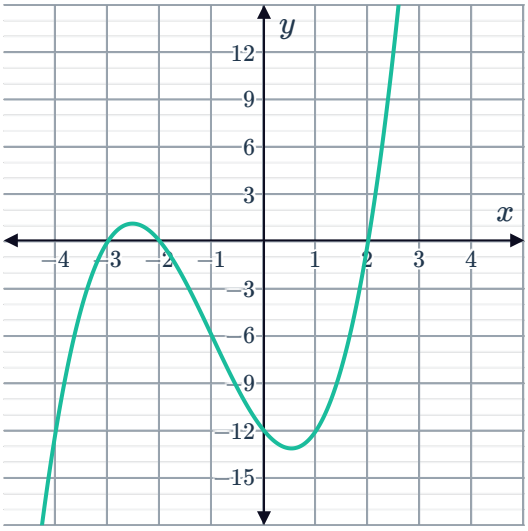
21

a

i $x = -3, -2, 2$

ii $y = -12$

iii

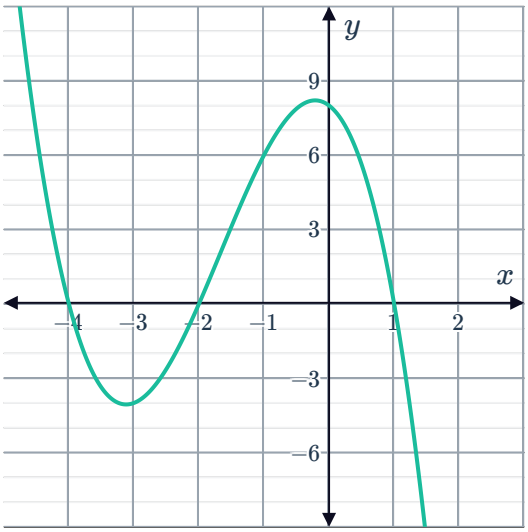


b

i $x = -4, -2, 1$

ii $y = 8$

iii



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

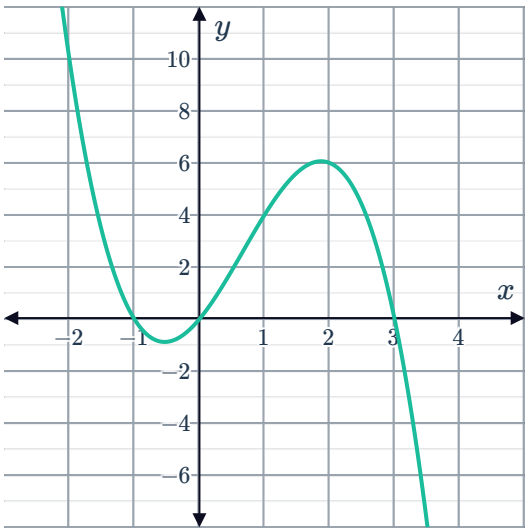
22

a $y = -x\,(x-3)\,(x+1)$

b $y = 0$

c $x = 0, 3, -1$

d



© ClipSAT: By Mr. Mohamed Abdallah +966597688647